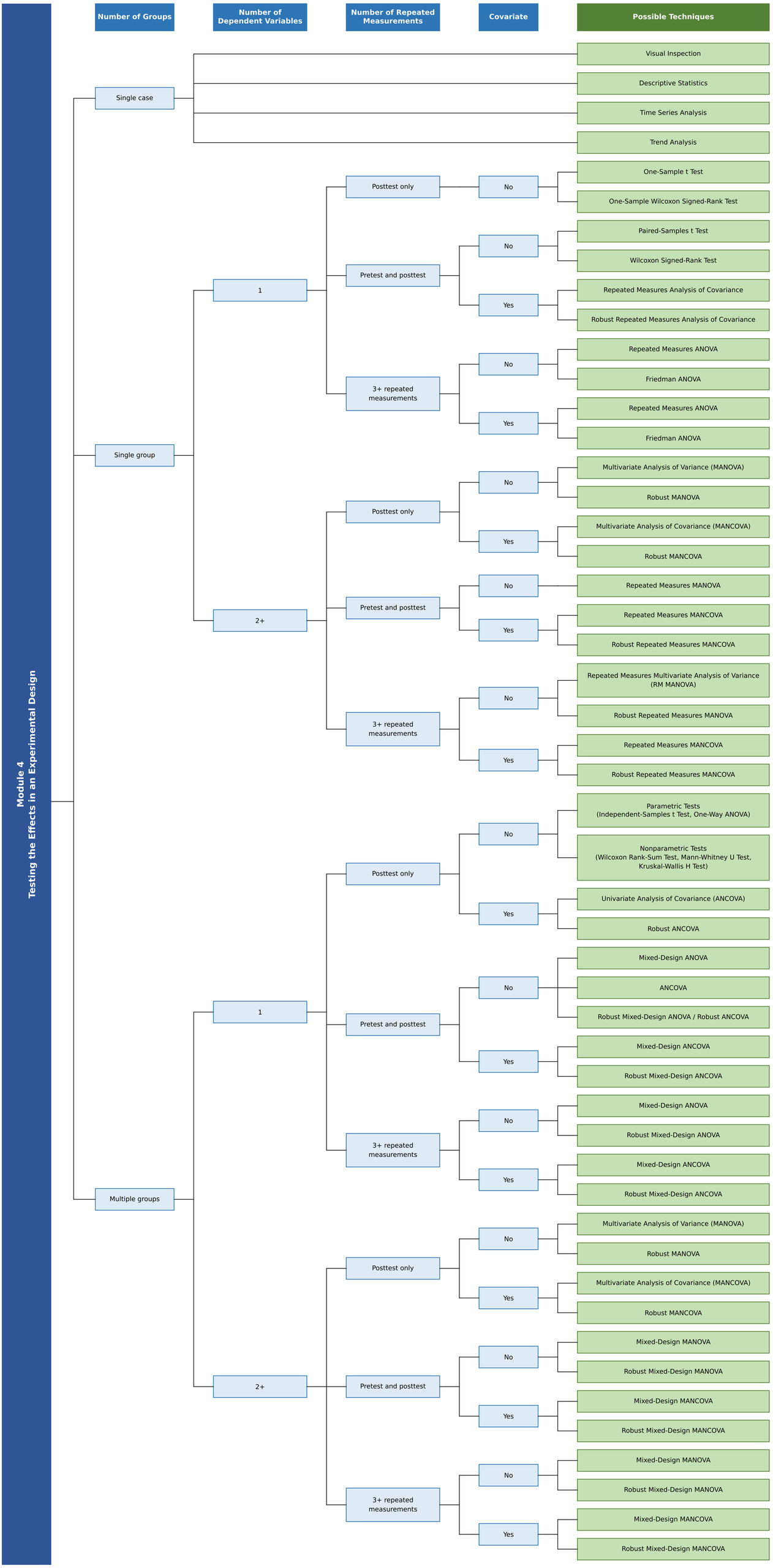
Module 4
Testing the Effects in an Experimental Design
Number of Groups
Number of Dependent Variables
Number of Repeated Measurements
Covariate
Possible Techniques
Single case
Visual Inspection
Descriptive Statistics
Time Series Analysis
Trend Analysis
Single group
1
Posttest only
No
One-Sample t Test
One-Sample Wilcoxon Signed-Rank Test
Pretest and posttest
No
Paired-Samples t Test
Wilcoxon Signed-Rank Test
Yes
Repeated Measures Analysis of Covariance
Robust Repeated Measures Analysis of Covariance
3+ repeated measurements
No
Repeated Measures ANOVA
Friedman ANOVA
Yes
Repeated Measures ANOVA
Friedman ANOVA
2+
Posttest only
No
Multivariate Analysis of Variance (MANOVA)
Robust MANOVA
Yes
Multivariate Analysis of Covariance (MANCOVA)
Robust MANCOVA
Pretest and posttest
No
Repeated Measures MANOVA
Yes
Repeated Measures MANCOVA
Robust Repeated Measures MANCOVA
3+ repeated measurements
No
Repeated Measures Multivariate Analysis of Variance (RM MANOVA)
Robust Repeated Measures MANOVA
Yes
Repeated Measures MANCOVA
Robust Repeated Measures MANCOVA
Multiple groups
1
Posttest only
No
Parametric Tests (Independent-Samples t Test, One-Way ANOVA)
Nonparametric Tests (Wilcoxon Rank-Sum Test, Mann-Whitney U Test, Kruskal-Wallis H Test)
Yes
Univariate Analysis of Covariance (ANCOVA)
Robust ANCOVA
Pretest and posttest
No
Mixed-Design ANOVA
ANCOVA
Robust Mixed-Design ANOVA / Robust ANCOVA
Yes
Mixed-Design ANCOVA
Robust Mixed-Design ANCOVA
3+ repeated measurements
No
Mixed-Design ANOVA
Robust Mixed-Design ANOVA
Yes
Mixed-Design ANCOVA
Robust Mixed-Design ANCOVA
2+
Posttest only
No
Multivariate Analysis of Variance (MANOVA)
Robust MANOVA
Yes
Multivariate Analysis of Covariance (MANCOVA)
Robust MANCOVA
Pretest and posttest
No
Mixed-Design MANOVA
Robust Mixed-Design MANOVA
Yes
Mixed-Design MANCOVA
Robust Mixed-Design MANCOVA
3+ repeated measurements
No
Mixed-Design MANOVA
Robust Mixed-Design MANOVA
Yes
Mixed-Design MANCOVA
Robust Mixed-Design MANCOVA